

LEC 1

أسالة

MCQs – History and Basics of Deep Learning

1. Who proposed the perceptron in 1958?

- A) McCulloch & Pitts
- B) Rosenblatt ☒
- C) Widrow & Hoff
- D) Minsky & Papert

Answer: B) Rosenblatt

Explanation: Rosenblatt introduced the perceptron as a model for binary classification with trainable weights.

2. What is the primary function of a perceptron?

- A) Solve XOR problems
- B) Perform linear binary classification ☒
- C) Train deep neural networks
- D) Recognize multiple categories at once

Answer: B) Perform linear binary classification

Explanation: A perceptron outputs +1 or -1 depending on whether the weighted sum of inputs exceeds a threshold.

3. What limitation does a single-layer perceptron have?

- A) Cannot classify linearly separable data
- B) Cannot solve non-linear problems like XOR ☒
- C) Cannot update weights
- D) Cannot handle binary inputs

Answer: B) Cannot solve non-linear problems like XOR

Explanation: Single-layer perceptrons can only separate linearly separable data.

4. Which algorithm is used to train multi-layer perceptrons (MLPs)?

- A) Perceptron learning algorithm
- B) Gradient descent/backpropagation ☒
- C) XOR algorithm
- D) Neocognitron

Answer: B) Gradient descent/backpropagation

Explanation: Rosenblatt's algorithm cannot train intermediate layers; backpropagation allows training deep networks.

5. What year did Minsky and Papert publish "Perceptrons"?

- A) 1958
- B) 1960
- C) 1969 ☒
- D) 1974

Answer: C) 1969

Explanation: This book highlighted limitations of perceptrons, contributing to the first AI winter.

6. What is the main reason for the AI winter after 1969?

- A) Lack of computers
- B) Perceptrons could not solve XOR and hype was too high ☒
- C) No datasets available
- D) Neural networks became too accurate

Answer: B) Perceptrons could not solve XOR and hype was too high

Explanation: Funding decreased because initial results were overhyped and perceived as failures.

7. Who introduced the Adaline model?

- A) Rosenblatt
- B) Widrow & Hoff ☒
- C) LeCun
- D) Hinton

Answer: B) Widrow & Hoff

Explanation: Adaline (Adaptive Linear Neuron) uses linear regression ideas to update weights.

8. What does a perceptron output when the weighted sum of inputs plus bias is positive?

- A) -1
- B) 0
- C) +1 ☒
- D) Cannot predict

Answer: C) +1

Explanation: The perceptron threshold function outputs +1 if the sum exceeds 0, otherwise -1.

9. What is the purpose of a bias in a perceptron?

- A) Acts as a weight
- B) Shifts the decision boundary ☒
- C) Normalizes inputs
- D) Solves XOR

Answer: B) Shifts the decision boundary

Explanation: Bias allows the threshold to move, changing the linear separation.

10. How many weights are needed for a neural network with a 200x200 RGB image input and 500 outputs?

- A) 6,000
- B) 60,000
- C) 60,000,000 ☒
- D) 60,000,000,000

Answer: C) 60,000,000

Explanation: Each pixel has 3 color channels $\rightarrow 200 \times 200 \times 3 = 120,000$ inputs. For 500 outputs $\rightarrow 120,000 \times 500 = 60\text{M}$ weights.

11. Who is credited with the first formal backpropagation algorithm?

- A) Rumelhart, Hinton, Williams
- B) Werbos ☒
- C) Linnainmaa
- D) Fukushima

Answer: C) Linnainmaa (1970)

Explanation: Linnainmaa introduced the reverse-mode automatic differentiation, forming the basis for backpropagation.

12. Which network architecture solved the vanishing gradient issue for sequences?

- A) MLP
- B) CNN
- C) LSTM ☒
- D) Perceptron

Answer: C) LSTM

Explanation: Long Short-Term Memory networks (Hochreiter & Schmidhuber, 1997) can capture long-term dependencies.

13. Who developed convolutional networks for OCR in the late 1990s?

- A) LeCun, Bottou, Bengio, Haffner ☒
- B) Rumelhart, Hinton, Williams
- C) Rosenblatt
- D) Minsky & Papert

Answer: A) LeCun et al.

Explanation: CNNs were applied successfully to handwritten digit recognition (MNIST).

14. What year was AlexNet introduced?

- A) 2009
- B) 2012 ☒
- C) 2015
- D) 1998

Answer: B) 2012

Explanation: AlexNet (Krizhevsky et al.) reignited deep learning popularity using GPUs on ImageNet.

15. What is the significance of the XOR problem in neural networks?

- A) Shows that linear classifiers are sufficient
- B) Demonstrates need for multi-layer networks ☒
- C) Solves all AI problems
- D) Cannot be learned by any algorithm

Answer: B) Demonstrates need for multi-layer networks

Explanation: XOR is not linearly separable, requiring hidden layers or feature engineering.

16. Why couldn't Rosenblatt's perceptron algorithm train intermediate layers?

- A) Lack of computing power
- B) No ground truth labels for hidden neurons ☒
- C) Weights were fixed
- D) Only worked for images

Answer: B) No ground truth labels for hidden neurons

Explanation: Perceptron updates weights based on output errors; hidden layers don't have direct labels.

17. What type of inputs did McCulloch & Pitts neurons originally use?

- A) Continuous inputs
- B) Binary inputs ☒
- C) RGB images
- D) Real-valued vectors

Answer: B) Binary inputs

Explanation: Early neural models were binary threshold units without learning.

18. Which of the following was a result of the first AI winter?

- A) Neural networks became widely funded
- B) Funding decreased and research slowed ☒

- C) LSTM networks were invented
- D) ImageNet was created

Answer: B) Funding decreased and research slowed

Explanation: The AI winter lasted roughly 1969–1983 due to unmet expectations.

19. What did Minsky and Papert highlight about perceptrons?

- A) They can solve XOR
- B) Limitations on linearly inseparable problems ☒
- C) Backpropagation works perfectly
- D) Multi-layer perceptrons are unnecessary

Answer: B) Limitations on linearly inseparable problems

Explanation: They showed single-layer perceptrons couldn't solve certain non-linear tasks.

20. Who proposed the concept of a machine that “senses, recognizes, remembers, and responds like a human mind”?

- A) Hinton
- B) Rosenblatt ☒
- C) LeCun
- D) Widrow

Answer: B) Rosenblatt

Explanation: Rosenblatt envisioned intelligent automata in 1958, motivating early AI research.

MCQs – Second AI Winter & Rise of Deep Learning

1. When did the second AI winter approximately occur?

- A) 1969–1983
- B) 1995–2006 ☒
- C) 2006–2012
- D) 1986–1995

Answer: B) 1995–2006

Explanation: During this period, traditional machine learning methods outperformed shallow neural networks, leading to reduced interest in deep learning.

2. Which of the following contributed to better performance than shallow neural networks during the second AI winter?

- A) Support vector machines (SVMs) ☒
- B) Perceptrons
- C) Single-layer MLPs
- D) Boltzmann machines

Answer: A) Support vector machines (SVMs)

Explanation: SVMs (Cortes & Vapnik, 1995) provided strong performance with solid mathematical foundations.

3. Who introduced Random Forests?

- A) Tin Kam Ho
- B) Breiman ☒
- C) Olshausen
- D) Hinton

Answer: B) Breiman

Explanation: Random Forests (2001) are ensemble methods combining multiple decision trees for better performance.

4. Which method is used for sparse coding?

- A) LSTM
- B) K-SVD ☒
- C) CNN
- D) DBN

Answer: B) K-SVD

Explanation: Sparse coding methods like K-SVD (Olshausen & Field, 1997) represent data efficiently using a small number of active components.

5. What is the purpose of manifold learning techniques like Isomap or LLE?

- A) Classify images
- B) Reduce dimensionality while preserving data structure ☒

- C) Train CNNs
- D) Solve XOR

Answer: B) Reduce dimensionality while preserving data structure

Explanation: Manifold learning uncovers lower-dimensional structures in high-dimensional data.

6. What event marked the “thaw” of the AI winter?

- A) Introduction of single-layer perceptrons
- B) Hinton & Salakhutdinov’s pretraining of deep networks in 2006 ☒
- C) McCulloch & Pitts neurons
- D) SVM invention

Answer: B) Hinton & Salakhutdinov’s pretraining of deep networks in 2006

Explanation: Layer-wise pretraining allowed deep networks to be trained effectively for the first time.

7. Deep Belief Networks (DBNs) are based on which model?

- A) Perceptrons
- B) Boltzmann Machines ☒
- C) LSTMs
- D) Decision Trees

Answer: B) Boltzmann Machines

Explanation: DBNs stack restricted Boltzmann machines (RBMs) and are pretrained layer by layer.

8. What problem did early neural networks suffer from before 2006?

- A) Lack of non-linearities
- B) Vanishing gradients ☒
- C) Too much data
- D) SVM competition

Answer: B) Vanishing gradients

Explanation: Gradients became very small in deep networks, making weight updates ineffective.

9. What was a major obstacle to training multi-layer perceptrons before the rise of deep learning?

- A) Overfitting and limited data ☒
- B) Lack of activation functions
- C) No backpropagation
- D) Too many layers

Answer: A) Overfitting and limited data

Explanation: Small datasets and limited computation prevented effective training of deep networks.

10. What is the main idea of layer-by-layer pretraining?

- A) Train all layers simultaneously
- B) Train one layer at a time, freezing previous layers ☒
- C) Skip hidden layers
- D) Use only one layer networks

Answer: B) Train one layer at a time, freezing previous layers

Explanation: Layer-wise training helps initialize weights sensibly before fine-tuning with backpropagation.

11. In layer-wise training, after training layer 1, what is done?

- A) Discard layer 1
- B) Freeze layer 1 and train layer 2 ☒
- C) Train all layers together
- D) Reset weights

Answer: B) Freeze layer 1 and train layer 2

Explanation: Freezing previously trained layers prevents them from being corrupted during the training of deeper layers.

12. Which year did Hinton & Salakhutdinov introduce pretraining for deep networks?

- A) 1997
- B) 2001

- C) 2006 ☒
- D) 2012

Answer: C) 2006

Explanation: This work reignited interest in deep neural networks by solving the training problem.

13. Which of these problems was not a challenge for early neural networks?

- A) Lack of processing power
- B) Overfitting
- C) Vanishing gradients
- D) Too many datasets ☒

Answer: D) Too many datasets

Explanation: Early neural networks struggled due to **lack** of data, not excess.

14. What is the advantage of multi-layer networks over single-layer networks?

- A) Easier training
- B) Ability to learn hierarchical and complex representations ☒
- C) Less computation
- D) No need for activation functions

Answer: B) Ability to learn hierarchical and complex representations

Explanation: Deeper networks can model complex, non-linear patterns better than shallow networks.

15. Which method is used to fine-tune all layers after layer-wise pretraining?

- A) Backpropagation ☒
- B) SVM
- C) LLE
- D) K-SVD

Answer: A) Backpropagation

Explanation: After layer-wise pretraining, backpropagation adjusts all weights jointly for optimal performance.

16. What was a typical number of hidden layers in early MLP experiments before deep learning?

- A) 1–2 hidden layers ☒
- B) 5–10 hidden layers
- C) 10–50 hidden layers
- D) No hidden layers

Answer: A) 1–2 hidden layers

Explanation: Early networks were shallow due to difficulty in training and lack of computational resources.

17. Why did deep learning become practical after 2006?

- A) Single-layer networks were enough
- B) Layer-wise pretraining + backpropagation + GPUs ☒
- C) Vanishing gradients disappeared automatically
- D) Datasets became smaller

Answer: B) Layer-wise pretraining + backpropagation + GPUs

Explanation: Combination of pretraining, better algorithms, and hardware enabled effective deep network training.

18. What is the benefit of freezing layers in deep learning?

- A) Prevent overfitting on previous layers ☒
- B) Make layers random
- C) Avoid using activation functions
- D) Reduce the number of neurons

Answer: A) Prevent overfitting on previous layers

Explanation: Freezing prevents weight updates from disrupting previously learned representations.

19. What was the main contribution of Hinton's 2006 work?

- A) Invented SVMs
- B) Demonstrated effective training of multi-layer networks via pretraining ☒
- C) Introduced Random Forests
- D) Solved XOR problem

Answer: B) Demonstrated effective training of multi-layer networks via pretraining

Explanation: This work showed deep networks could be trained efficiently with layer-wise pretraining.

20. Which deep learning challenge is directly addressed by layer-wise pretraining?

- A) Lack of datasets
- B) Vanishing gradients ☒
- C) Overfitting only
- D) Data augmentation

Answer: B) Vanishing gradients

Explanation: Layer-wise pretraining initializes weights in a way that reduces the vanishing gradient problem during full backpropagation.

MCQs – Deep Learning: Big Data, ImageNet, Modern Models

1. When was the ImageNet dataset first published?

- A) 2006
- B) 2009 ☒
- C) 2012
- D) 2015

Answer: B) 2009

Explanation: ImageNet (Deng et al., 2009) contains ~16M labeled images covering 100K WordNet terms.

2. How many categories does the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012 use?

- A) 100
- B) 1,000 ☒
- C) 10,000
- D) 100,000

Answer: B) 1,000

Explanation: ILSVRC 2012 used 1 million images in 1,000 categories for top-1 and top-5 error evaluation.

3. Approximately how many images are in the ImageNet-12 dataset?

- A) 130,000
- B) 1.3M ☒
- C) 13M
- D) 16M

Answer: B) 1.3M

Explanation: ImageNet-12 has ~1.3 million images (~150GB), commonly used for training deep networks.

4. Which deep learning model won the ILSVRC 2012 challenge?

- A) ResNet
- B) AlexNet ☒
- C) LeNet
- D) VGG

Answer: B) AlexNet

Explanation: AlexNet (Krizhevsky, Sutskever & Hinton) showed that deep CNNs could outperform traditional methods.

5. AlexNet had more weights than what?

- A) Training layers
- B) Samples in the dataset ☒
- C) GPU memory
- D) Categories

Answer: B) Samples in the dataset

Explanation: AlexNet had millions of parameters, more than the 1M images it was trained on.

6. What major factor enabled deep learning success around 2012?

- A) Smaller datasets
- B) Better hardware, more data, better algorithms ☒
- C) Less powerful GPUs
- D) Shallow networks

Answer: B) Better hardware, more data, better algorithms

Explanation: GPUs, large datasets (like ImageNet), and improved training algorithms made deep networks practical.

7. What year was ResNet (154 layers) introduced?

A) 2012

B) 2015 ☒

C) 2017

D) 2020

Answer: B) 2015

Explanation: ResNet introduced skip connections to train very deep networks effectively.

8. Which major AI milestone was achieved by DeepMind in 2015?

A) ImageNet challenge

B) Playing Go with AI ☒

C) GPT-3 release

D) AlexNet development

Answer: B) Playing Go with AI

Explanation: DeepMind's AlphaGo beat professional Go players, showcasing reinforcement learning.

9. Which dataset contains 1,000 classes and ~1.3M images?

A) CIFAR-10

B) MNIST

C) ImageNet-12 ☒

D) COCO

Answer: C) ImageNet-12

Explanation: This large dataset enabled training of deep CNNs like AlexNet.

10. GPT-3 has approximately how many parameters?

A) 354M

B) 175B ☒

- C) 6144
- D) 1M

Answer: B) 175B

Explanation: GPT-3 is a very large language model requiring extensive compute (~1500 GPUs for 2 months).

11. Training GPT-3 roughly costs?

- A) \$2K
- B) \$350K
- C) \$3M ☒
- D) \$25M

Answer: C) \$3M

Explanation: Training GPT-3 on 175B parameters was extremely resource-intensive.

12. PaLM (Google) required how many TPUs and approximate cost?

- A) 1,000 TPUs, \$3M
- B) 6,144 TPUs, \$25M ☒
- C) 100 TPUs, \$2K
- D) 500 TPUs, \$350K

Answer: B) 6,144 TPUs, \$25M

Explanation: Large-scale models like PaLM need massive computational resources.

13. What energy consumption was estimated for PaLM training?

- A) 1 kWh
- B) 3,200 kWh
- C) 3.2 million kWh ☒
- D) 25 kWh

Answer: C) 3.2 million kWh

Explanation: Equivalent to ~1,000 households consuming electricity for a year.

14. Which model has ~354M parameters?

- A) GPT-3

- B) BERT ☒
- C) AlexNet
- D) ResNet

Answer: B) BERT

Explanation: BERT (Bidirectional Encoder Representations from Transformers) is smaller than GPT-3 but large for NLP.

15. What is a major challenge for training deep learning models?

- A) Lack of data ☒
- B) Too few parameters
- C) No datasets available
- D) Overfitting solved automatically

Answer: A) Lack of data

Explanation: Deep networks need massive datasets to avoid overfitting.

16. What is the “wow” factor of deep learning today?

- A) Small datasets
- B) Ability to perform object recognition, language generation, image synthesis ☒
- C) Only OCR
- D) Only simple classification

Answer: B) Ability to perform object recognition, language generation, image synthesis

Explanation: Deep learning scales to many domains, including NLP, vision, and art.

17. Which model is associated with playing Atari games with reinforcement learning?

- A) AlexNet
- B) Deep Q-Network (DQN) ☒
- C) ResNet
- D) GPT

Answer: B) Deep Q-Network (DQN)

Explanation: Mnih et al., 2013, used deep RL for learning Atari gameplay from raw pixels.

18. SegNet is primarily used for what?

- A) Language translation
- B) Image segmentation ☒
- C) Text classification
- D) Speech recognition

Answer: B) Image segmentation

Explanation: SegNet is an encoder-decoder CNN architecture for pixel-wise segmentation.

19. What is NeuralTalk used for?

- A) Image classification
- B) Video classification
- C) Image captioning ☒
- D) Game playing

Answer: C) Image captioning

Explanation: Karpathy et al., 2014, used CNN+RNN for large-scale video/image captioning.

20. DALL-E 2 (2022) is designed for?

- A) Object recognition
- B) Generating images from text prompts ☒
- C) Speech recognition
- D) Playing Go

Answer: B) Generating images from text prompts

Explanation: DALL-E 2 is a generative model for creative image synthesis.

21. Why did deep learning become practical only after 2009?

- A) Better algorithms only
- B) Bigger datasets, better hardware, improved algorithms ☒
- C) Shallow networks were enough
- D) Vanishing gradients disappeared naturally

Answer: B) Bigger datasets, better hardware, improved algorithms

Explanation: All three factors were necessary for training deep networks effectively.

22. What is ILSVRC top-5 error?

- A) Number of images misclassified ☒
- B) Percentage where correct label is in top 5 predictions
- C) Accuracy of shallow networks
- D) SVM benchmark

Answer: B) Percentage where correct label is in top 5 predictions

Explanation: Used as a standard metric for large-scale image classification.

23. Which deep learning breakthrough is linked to 2012?

- A) ResNet
- B) AlexNet ☒
- C) BERT
- D) SegNet

Answer: B) AlexNet

Explanation: Demonstrated the power of deep CNNs on ImageNet.

24. What does ResNet introduce to enable very deep networks?

- A) Dropout
- B) Skip (residual) connections ☒
- C) LSTMs
- D) Layer freezing

Answer: B) Skip (residual) connections

Explanation: Helps gradients flow through very deep networks.

25. Which factor is NOT a reason deep learning is data hungry?

- A) Large number of parameters
- B) Need for high generalization
- C) Small-scale images only ☒
- D) Deep networks require diverse examples

Answer: C) Small-scale images only

Explanation: Deep learning requires **large and diverse datasets**, not small datasets.

26. GPT-3 required training on approximately how many GPUs?

- A) 10
- B) 100
- C) 1,500 ☒
- D) 6,144

Answer: C) 1,500

Explanation: Massive parallel compute was needed for training 175B parameter model.

27. Which of the following is a milestone of deep learning in games?

- A) SegNet
- B) AlphaGo ☒
- C) DALL-E 2
- D) BERT

Answer: B) AlphaGo

Explanation: Deep reinforcement learning allowed AI to beat top Go players.

28. ImageNet helped advance deep learning primarily because?

- A) It was small
- B) It provided large-scale labeled image datasets ☒
- C) It was free of errors
- D) It used text data

Answer: B) It provided large-scale labeled image datasets

Explanation: Large datasets are essential for training deep CNNs.

29. Deep learning is considered “big data hungry” because?

- A) Neural networks are small
- B) Training requires millions of examples to avoid overfitting ☒
- C) Datasets are irrelevant
- D) Algorithms are outdated

Answer: B) Training requires millions of examples to avoid overfitting

Explanation: Deep models with millions of parameters need massive datasets to generalize.

30. The deep learning golden era roughly starts from?

A) 2006 ☒

B) 1995

C) 2012

D) 2015

Answer: A) 2006

Explanation: Layer-wise pretraining and improved algorithms mark the modern rise of deep learning.

MCQs – Modern Deep Learning & AI Applications

1. What is the purpose of <https://betterimagesofai.org/>?

A) Train neural networks

B) Depict AI accurately and critically ☒

C) Generate AI music

D) Benchmark ImageNet

Answer: B) Depict AI accurately and critically

Explanation: The site aims to reduce misleading hype and show AI realistically.

2. DeepMind's AlphaGo solves games like Go using:

A) Pure brute-force search

B) Neural networks + tree search ☒

C) Random moves

D) Preprogrammed rules

Answer: B) Neural networks + tree search

Explanation: AlphaGo combines deep RL and Monte Carlo tree search to handle huge game complexity.

3. In Go, the approximate number of legal game sequences is:

- A) 10^6
- B) 10^{80}
- C) Much larger than 10^{80} ☒
- D) 10^{50}

Answer: C) Much larger than 10^{80}

Explanation: The search space for Go is astronomically large, exceeding even the number of atoms in the universe.

4. DeepMind also applies AI to which scientific domain?

- A) Protein folding ☒
- B) Chess only
- C) Simple board games
- D) ImageNet

Answer: A) Protein folding

Explanation: AI helps predict protein structures, weather, and other complex scientific problems.

5. Which paper introduced hierarchical text-conditional image generation with CLIP latents?

- A) GPT-3
- B) Ramesh et al., 2022 ☒
- C) AlexNet
- D) SegNet

Answer: B) Ramesh et al., 2022

Explanation: This research enabled multi-modal models combining vision and language for image generation.

6. Approximately how many images were used to train the CLIP-based image generation encoder?

- A) 1M
- B) 16M

C) 650M ☒

D) 1B

Answer: C) 650M

Explanation: Large-scale datasets enable vision-text multi-modal models to generalize well.

7. How much raw text data was used for GPT pretraining before filtering?

A) 570GB

B) 45TB ☒

C) 100GB

D) 1TB

Answer: B) 45TB

Explanation: GPT models pretrain on massive amounts of internet text to learn language patterns.

8. After filtering, GPT training text data was reduced to approximately:

A) 45TB

B) 1TB

C) 570GB ☒

D) 100GB

Answer: C) 570GB

Explanation: Data cleaning removes low-quality, redundant, or irrelevant content.

9. Which historical figure predicted AI could compose music using an analytical engine?

A) Alan Turing

B) Ada Lovelace ☒

C) Charles Babbage

D) Marvin Minsky

Answer: B) Ada Lovelace

Explanation: Lovelace theorized that computers could manipulate symbols beyond numbers, including music.

10. Mubert AI demonstrates AI in which domain?

- A) Text generation
- B) Music generation ☒
- C) Robotics
- D) Image captioning

Answer: B) Music generation

Explanation: Mubert converts text prompts into AI-generated music.

11. RMA (Rapid Motor Adaptation) applies deep learning to:

- A) Vision datasets
- B) Language translation
- C) Legged robots' joint control ☒
- D) Board games

Answer: C) Legged robots' joint control

Explanation: AI can now help robots learn speed, force, and direction control.

12. Before deep learning in robotics, joint movement was primarily:

- A) Learned automatically
- B) Preprogrammed manually ☒
- C) Random
- D) Controlled by vision only

Answer: B) Preprogrammed manually

Explanation: Deep learning allows adaptive motor control without exhaustive programming.

13. To stay up-to-date with deep learning research, one recommended source is:

- A) ImportAI ☒
- B) Facebook
- C) YouTube only
- D) Twitter memes

Answer: A) ImportAI

Explanation: ImportAI and newsletters like The Batch and Deep Learning Weekly provide curated updates.

14. The Batch, ImportAI, and Deep Learning Weekly are examples of:

- A) Deep learning frameworks
- B) Weekly newsletters for AI research ☒
- C) Datasets
- D) Robotics controllers

Answer: B) Weekly newsletters for AI research

Explanation: These help researchers keep track of the fast-moving field.

15. Craiyon is used for:

- A) Text-to-image generation ☒
- B) Language modeling
- C) Game-playing AI
- D) Image classification

Answer: A) Text-to-image generation

Explanation: Craiyon (formerly DALL-E mini) generates images from text prompts.

16. HuggingFace notebooks for stable diffusion allow:

- A) Preprocessing text only
- B) Running text-to-image AI locally or on Colab ☒
- C) Robot motor control
- D) Video classification

Answer: B) Running text-to-image AI locally or on Colab

Explanation: Provides interactive experimentation with generative models.

17. HuggingFace GPT-Neo 1.3B allows:

- A) Small language model experimentation ☒
- B) Image classification

- C) Reinforcement learning
- D) Protein folding

Answer: A) Small language model experimentation

Explanation: GPT-Neo is an open-source transformer for NLP research.

18. What is emphasized as important when learning deep learning?

- A) Only theory
- B) Only practical skills
- C) Both theory and practical understanding ☒
- D) Only memorizing papers

Answer: C) Both theory and practical understanding

Explanation: Understanding fundamentals is key before exploring research.

19. Which area is not explicitly mentioned as a modern AI application in these slides?

- A) Image generation
- B) Music generation
- C) Robotics
- D) Quantum computing ☒

Answer: D) Quantum computing

Explanation: Slides focus on vision, language, music, and robotics, not quantum AI.

20. The recommended approach for keeping up with AI research includes:

- A) Reading old textbooks only
- B) Reading exciting papers regularly ☒
- C) Ignoring news
- D) Only programming

Answer: B) Reading exciting papers regularly

Explanation: Engaging with current research helps build understanding and intuition.

ملخص

Frank Rosenblatt و Perceptrons (1958)

- ، وكان هدفه يعمل حاجة شبه المخ البني آدم: يحس، يعرف، يفكر **Perceptron** روزنبلات عمل نموذج اسمه ويرد.
- ، لو الناتج أكبر من صفر يبقى 1، لو أقل **bias** النموذج بياخد مدخلات مستمرة ويضربها في أوزان ويجمعهم مع .يبقى -1.
- .ده كان أول نموذج عملي للتعلم الآلي.

Perceptron تدريب الـ

- فيه خوارزمية للتعلم:
 1. ابدأ بالأوزان عشوائية.
 2. (بتاعها label صورة و) خذ مثال جديد.
 3. احسب الناتج.
 4. لو الناتج غلط، عدل الأوزان:
 - لو لازم يطلع 1 وطلع -1 → زود الوزن
 - لو لازم يطلع -1 وطلع 1 → نقص الوزن
 5. كرر العملية لحد ما الشبكة تبقى شاطرة.

من إخراج واحد لكذا إخراج

- .(أو 1-1) الأول كان بيشتغل بس على قرار ثنائي Perceptron الـ.
- .لو عايزين نصنف حاجات أكثر (مثلاً 500 فئة لصور)، نعمل شبكة عصبية بأكثر من إخراج، كل فئة لها وزنها.
- .(مليون 60 ألوان، و 500 فئة → الحساب يطلع عدد أوزان ضخمة جدًا (حوالي ، 2003x مثال: صورة 200

Perceptron والـ XOR مشكلة

- .XOR واحد لير مش بيقدّر يحل مشاكل غير خطية زي Perceptron الـ.
- .العادي مش هيقدر Perceptron ، الـ XOR يعني لو عندنا مدخلين وكل مدخل ممكن 0 أو 1، ومطلوب نطلع .يفرق بينهم بخط مستقيم.

الحل: Multi-Layer Perceptrons (MLP)

- إخراج طبقة يكون مدخل للطبقة اللي بعدها: نعمل طبقات أكثر.
- sigmoid بين كل طبقة والثانية نستخدم تفعيلات غير خطية زي.
- وأي مشكلة معقدة XOR كده الشبكة تقدر تحل.
- لكل طبقة داخلية، وده مش "ground truth" بس مشكلة: خوارزمية روزنبلات مش هتشتغل هنا، لأننا محتاجين متاح.

AI Winter (1969-1983)

- مش قادر يحل Perceptron ، الناس خافت وقالت: "لو Perceptron عن الـ Minsky & Papert بعد كتاب "، ليه نهتم؟ XOR
- AI Winter. التمويل توقف → حصل
- مع ذلك، اكتشفوا حاجات مهمة في الفترة دي:
 - Backpropagation: MLP خوارزمية لتعلم الـ
 - Recurrent Networks: لمعالجة تسلسلات طولها متغير
 - CNNs: Neocognitron لتصنيف الصور

ملخص سريع للتسلسل الزمني

السنة	الحدث
1958	Perceptron ،Rosenblatt
1960	Adaline ،Widrow & Hoff
1969	Perceptron ينتقدوا الـ Minsky & Papert
1970-1986	يظهر تدريجياً Backpropagation
1997	LSTM ،Hochreiter & Schmidhuber
1998-2012	على الصور CNNs تطبيق (LeCun ،Hinton ،AlexNet)

السنة	الحدث
2015	GO يتفوق في DeepMind ،ResNet

تمام، هشرحلك النقاط دي بالعامية المصرية وبطريقة بسيطة وملخصة بحيث تقدر تفهمي كل حاجة بسرعة

1. الثانية “AI winter” (1995 – 2006)

- (اللي كان 1969 – 1983)، ظهر ثاني فترة ركود للذكاء الاصطناعي AI winter بعد أول
- Machine Learning كانت موجودة لكن الناس عملت أساليب ثانية بالـ Neural Networks السبب: شبكات الـ وأثبتت نفس الدقة أو أحسن
- أمثلة على الأساليب اللي ظهرت
 - **Kernel methods**: أساليب رياضية لتحويل البيانات عشان نسهل التصنيف
 - **SVM (Support Vector Machines)**: كوريس وفابنك 1995، طريقة قوية للتصنيف
 - **Decision Trees & Random Forests**: Random Forest (Breiman, 2001) شجرة قرار، و مجموعات أشجار لنتائج أفضل
 - **Manifold learning** (~2000): LTSA و LLE و Isomap طرق لتقليل أبعاد البيانات زي
 - **Sparse coding** (Olshausen & Field, 1997): طريقة لتمثيل البيانات بطريقة أقل كثافة، زي LASSO و K-SVD.

الثانية كانت أسهل وأدق في مشاكل كثير ML مش كانت الأفضل، وطرق NNS في الفترة دي، الـ: **الملخص**

2. Deep Learning (2006 – present) صعود

- **layer by layer** اكتشفوا طريقة لتدريب شبكات كبيرة Hinton و Salakhutdinov في 2006،
- **backprop** يعني بدل ما تحاول تدريب كل الشبكة مرة واحدة، تدريب كل طبقة لوحدها وبعدين تضبطها بالـ
- **Boltzmann Machines** المبنية على **Deep Belief Networks (DBNs)** أسلوبهم أطلق

بدأنا نقدر نعمل شبكات أعمق وأدق لأننا لقينا طريقة لتسهيل التدريب: **الملخص**

3. Neural Networks قبل 10 سنين

- مشاكل:

- الكمبيوترات كانت ضعيفة: **processing power** قليلة الـ
- كبيرة datasets ما كانش فيه: **قلة البيانات**
- الشبكات كانت بتحفظ على البيانات ومش بتتعلم كويس: **Overfitting**
- **Vanishing gradients**: لما تحاول تدرب شبكة كبيرة، الأوزان في الطبقات البعيدة بتتوقف عن التحديث.
- النتيجة: تدريب شبكات كبيرة ما كانش مفيد كثير

كانت غالبًا أفضل اللي نقدر نعمله وقتها hidden layers شبكات 1-2: **الملخص**

4. دلوقتي Neural Networks

- الكبيرة، بقى ممكن ندرب شبكات عميقة بسهولة أكثر **Datasets** و **GPU** مشاكل كثير زي قبل، لكن بفضل الـ
 - بقى أسهل، وممكن نحقق فوائد الشبكات العميقة مع سهولة التدريب multi-layer networks تدريب
- التكنولوجيا الحديثة خلّيت الشبكات العميقة عملية أكثر من زمان: **الملخص**
-

5. layer-by-layer طريقة التدريب

- بدل ما تدرب كل الشبكة مرة واحدة، تدرب الطبقة الأولى الأول
 - وتدرب الطبقة اللي بعدها (freeze) بتاعتها weights بعد ما تتعلم كويس، تثبت
 - كده كل طبقة تتعلم أفضل وتمثل بياناتها بطريقة أحسن
 - مثال:
 - Train layer 1 → Freeze layer 1 → Train layer 2 → Freeze layer 2 → Train layer 3
- gradient. كل طبقة بتتعلم لوحدها عشان التدريب يبقى أسهل ونتجنب مشاكل: **الملخص**
-

تمام! هاشرحلك كل النقاط اللي انت بعتالنا بالعامية المصرية وبشكل **مختصر وسهل** بحيث تفهم الصورة الكبيرة عن تاريخ: من أول الأيام لحد النهاردة **Deep Learning** وتطور الـ

1. ImageNet – سنة 2009

- وفريقه Deng بواسطة **ImageNet** كبير اسمه **dataset** في 2009 اتعمل
- **WordNet** مليون صورة مصنفة على 100 ألف كلمة من 16 فيه حوالي

- فيها 1.3 مليون صورة و 1,000 فئة **ImageNet-12** النسخة الأكثر استخدامًا دلوقتي
 - الهدف: نخلي الموديلات تتعلم تمييز الكائنات في الصور
-

2. ImageNet Challenge (ILSVRC)

- مليون صورة و 1,000 فئة 1 مسابقة لتقييم النماذج على
 - **Top-1 error** و **Top-5 error** بيقيسوا الأداء بـ
 - كانوا بيشاركوا فيها **non-CNN** أو **CNN** سواء الموديلات كانت
-

3. AlexNet – 2012

- AlexNet فاز بمسابقة ImageNet 2012.
 - فيه عدد وزن أكثر من عدد العينات اللي اتدرب عليها
 - على الصور الكبيرة deep learning وفتح الطريق للـ CNN عمل ثورة في الـ
-

4. نجح؟ Deep Learning ليه دلوقتي

- كانت بتحتاج بيانات قليلة، وعملياتها بطيئة neural networks قبل كده: الـ
 - دلوقتي:
 1. **Hardware** – أحسن GPUs – سريعة جدًا
 2. ImageNet زي – بيانات كبيرة
 3. CNNs و backprop زي – خوارزميات أفضل
-

5. الموديلات الكبيرة اليوم

- BERT: 354 ألف دولار تقريبًا 2 تكلفة تقريبًا
- RoBERTa: 1000 GPU احتاج 350 ألف دولار
- GPT-3: 175 مليون دولار لتدريبه شهرين
- PaLM: 6144 TPU 3.2 مليون دولار، واستهلك طاقة ~ 3.2
- كبيرة بتحتاج موارد هائلة DL ده يوريك قد إيه موديلات

6. دلوقتي Deep Learning أهمية

- h5-index مرتبة في أفضل المجالات العلمية من حيث: أصبح من أهم المجالات العلمية
- ، video analysis ، robotics ، NLP ، كمان في computer vision مش بس للـ deep learning الـ arts ، games...

7. Deep Learning التطور الزمني السريع للـ

- 1958: Rosenblatt – Perceptrons.
- 1960: Widrow & Hoff – Adaline.
- 1969: Minsky & Papert – اكتشفوا مشاكل perceptrons.
- 1970–1986: Backpropagation ، LSTM اتطور ،
- 1998: LeCun & CNNs – OCR.
- 2006: Hinton & pretraining – Deep Belief Networks.
- 2009: ImageNet – big data.
- 2012: AlexNet – CNN on ImageNet.
- 2015: ResNet و DeepMind GO.
- 2020+: GPT-3, DALL-E, PaLM – موديلات ضخمة جداً

8. Deep Learning في البحث العلمي

- علشان تبدأ في البحث:
 1. فهم الأساسيات كويس (نظرية + عملي)
 2. اقرأ أوراق بحثية – في الأول هتبقى صعبة، مع الوقت هتفهم أكثر
 3. ، جرب موديلات مختلفة datasets التجربة مهمة جداً – لعب مع

9. عملية Deep Learning تطبيقات

- **Atari games** – Deep Reinforcement Learning (Mnih 2013).

- **SegNet** – CNN encoder-decoder للـ segmentation.
 - **Video classification** – Karpathy 2014.
 - **الـ arts** – DL كمان في الفن يستخدم!
-

10. الملخص الكبير.

- سريع، وخوارزميات ذكية hardware بيانات كبيرة،: **النهاردة شغال بسبب Deep Learning الـ**
 - الموديلات الحديثة كبيرة جدًا وبتحتاج موارد هائلة، لكن قدرتها على التعلم والتطبيقات عمليًا مذهلة.
 - بتمثل تطور تدريجي في فهم وتدريب الشبكات DALL-E و GPT-3 لحد Perceptron كل التطورات من العصبية.
-

تمام! هشرحلك النقط دي كلها بالعامية المصرية وبطريقة مختصرة وبسيطة عشان تفهم الصورة كلها بسرعة

1. Depictions of AI (slide 52-53)

- حقيقية مش مبالغ فيها AI بيوريك صور [Better Images of AI](#) فيه موقع اسمه
 - ، مهم تبقى ناقد ومتفكر قبل ما تصدق كل حاجة “AI hype” الفكرة: الصحافة ساعات بتضخم الموضوع أو تعمل عن الذكاء الاصطناعي.
-

2. AI beyond human capacity (slide 53-54)

- AlphaGo وورقة DeepMind: مثال كبير
 - أو الشطرنج، لو تحسب كل الاحتمالات يبقى الرقم خرافي (أكثر من عدد ذرات الكون) Go الألعاب الكبيرة زي
 - ، باستخدام الشبكات العصبية exhaustive search بس الذكاء الاصطناعي ممكن يحلها بطريقة ذكية، مش tree search.
 - دلوقتي بيشتغلوا على حاجات أكبر زي توقع البروتينات و التنبؤ بالطقس و كمان مشاكل علمية ثانية
-

3. Vision-text multi-modal learning (slide 54-55)

- فيه أبحاث دلوقتي بتخلي الموديلات تفهم صور ونصوص مع بعض
- CLIP Latents ، أو الورقة بتاعة OpenAI من DALL-E 2: مثال

- **650 مليون صورة** عشان تدريب الموديل ده، استخدموا حوالي
-

4. Generative pretraining (slide 55-56)

- OpenAI من *Language Models are Few-Shot Learners*: الورقة الشهيرة
 - **تيرا بايت نصوص من الإنترنت قبل الفترة** (بعدين بقي 570 جيجا) **45** استخدموا
 - **task** تتعلم من كمية ضخمة جدًا من البيانات قبل ما نديها أي GPT الفكرة: الموديلات الكبيرة زي
-

5. Music from AI (slide 56)

- يقدر يعمل **موسيقى AI** دلوقتي
 - **محول النصوص لموسيقى تلقائيًا** [Mubert AI](#): مثال
 - يعني مش بس صور ونصوص، كمان فن وموسيقى ممكن تتولد بالذكاء الاصطناعي
-

6. Ada Lovelace & AI pioneer (slide 56-57)

- Ada Lovelace ممكن الكمبيوتر يولد **”موسيقى علمية ومعقدة**
 - هي كانت من أوائل الناس اللي فكرت في فكرة إن الآلة ممكن تعمل أكثر من مجرد أرقام
-

7. Deep Learning in robotics (slide 57-58)

- بقي بيشتغل كمان في الروبوتات AI
 - **بيركز على تعليم الروبوت يتحرك صح** (سرعة، قوة، اتجاه) – RMA (Rapid Motor Adaptation): مثال
 - قبل كده، كان كل حركة لازم تتبرمج يدوي، دلوقتي الشبكات العصبية بتتعلم تتحرك لوحدها
-

8. Keeping up with research (slide 58-59)

- المجال بيتطور بسرعة رهيبه
- عشان تتابع الجديد، فيه نشرات أسبوعية
 - *The Batch* و *ImportAI*

- *Deep Learning Weekly*
 - *NLP News*
 - دي هتخليك متابع كل الجديد من غير ما تضيع وقت
-

9. Conclusion / summary (slide 59-60)

- ...وانتشر لكل المجالات: نصوص، موسيقى، روبوتات (Vision) الذكاء الاصطناعي والتعلم العميق بدأ من الرؤية
 - الحجم الكبير للبيانات مهم، بس مش الحل لكل حاجة
 - هدف الكورس: يعطيك الأساس عشان تقدر تتعلم وتتطور لوحدهك
-

10. Extra resources / if there's time (slide 60)

- AI: مواقع تجريبية أو أدوات تلعب بيها
 - توليد صور من نصوص – [Craiyon](#)
 - توليد صور – [Stable Diffusion Colab](#)
 - موديل لغة كبير تجرب عليه نصوص – [GPT-Neo 1.3B](#)
-

💡 الخلاصة بالعامية:

- ...مش بس صور وخوارزميات، ده بيشتغل في نصوص، صور، موسيقى، روبوتات AI
- أهم حاجة تبقى ناقد وواقعي، وتتابع الأبحاث الجديدة
- دلوقتي كل حاجة بتكبر: الموديلات أكبر، البيانات أكبر، الأجهزة أسرع، والتطبيقات واسعة جدًا